

PCT/PCM 7: Shape Surgery — Solutions

4: Interpret the Figure

Opening removes the small foreground speckles, while closing fills the holes in the objects. The order matters because opening performs erosion followed by dilation and therefore removes small foreground details, whereas closing performs dilation followed by erosion and fills small holes or gaps.

5: Erosion Calculation

A solid 10×10 foreground square is eroded with a 3×3 square structuring element (SE).

Output size = $(10 - 3 + 1) \times (10 - 3 + 1) = 8 \times 8$.

Output area = 64 pixels.

Original area = 100 pixels.

Area lost = $100 - 64 = 36$ pixels.

A 2-pixel-wide line is completely removed because it is narrower than the 3×3 SE.

6: Prediction

No. An object smaller than a 5×5 SE cannot survive opening because the initial erosion removes it, and the subsequent dilation cannot restore an object that was completely eliminated.